



UnionPay Integrated Circuit Card Specifications— Auxiliary Specifications

Part IX UnionPay Integrated Circuit Card Products Applications Guide



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1 Application Scope

This Part applies to all UPI Participants.

2 Division of UnionPay Integrated Circuit Card Functions/Products and Transaction Path

2.1 Division of Functions of UnionPay Integrated Circuit Card

2.1.1 Division based on Transaction Authorization Manner

At present, the function of the UnionPay Integrated Circuit Card is divided into offline function and online function according to the transaction authorization manner. The offline function that supports offline payment transactions is called “E-Cash” function; the online function that supports online payment transactions is called “Debit/Credit” function.

2.1.2 Contact and Contactless Path Division

The UnionPay Integrated Circuit Card transactions can also be divided into contact type and contactless type according to the transaction communication manner. The contact transactions require the UnionPay Integrated Circuit Card to have physical contact with the reader in order to complete the read-write operations; while the contactless transactions require the UnionPay Integrated Circuit Card to interact with the reader via wireless signals to complete the same.

The following functions may be combined according to the transaction communication manner and the authorization manner.

Table 1: Function Combinations

	Contact	Contactless
Online (Debit/Credit)	A: contact debit/credit	C: contactless debit/credit
Offline (E-Cash)	B: contact E-Cash	D: contactless E-Cash

2.2 Design of Financial IC Card Products

To promote the UnionPay Integrated Circuit Card business and expand the market, the UPI has designed two major product lines based on different function combinations of the UnionPay Integrated Circuit Card.

1. **“Debit/Credit Products”**: these products are designed to adapt to the habit of using traditional magstripe cards. They only support the contact online debit/credit function (refer to A in Table 1).
2. **“Quick Pass” products based on contactless applications**: these products may not only expand the application scope of traditional bank cards by supporting quick low-value payment, but also meet the requirements for mobile proximity payment and payment via devices without the contact

communication function, such as special shaped cards. Therefore, the "Quick Pass" products may simultaneously support the debit/credit function and the E-Cash function, that is to say, they can not only support online payments (refer to C in Table 1), but also support offline payments (refer to D in Table 1).

Due to limited applications of the contact E-Cash function, there have been no independent products developed so far (refer to B in Table 1).

2.3 Transaction Process of UnionPay Integrated Circuit Card (Path)

2.3.1 Main Transaction Process of UnionPay Integrated Circuit Card

According to the interaction path with the terminal, the transaction path of IC cards may be divided into 3 types, namely, the standard debit/credit process, contactless quick process (in short, qUICS) and low-value payment process based on the debit/credit applications.

2.3.1.1 Standard Debit/Credit Process

The standard debit/credit process is generally called complete process, which is consistent with the complete business process defined in the *UnionPay Integrated Circuit Card Specifications—Basic Specifications Part 1: Debit/Credit Application Overview* and support all functions therein. This process generally requires longer time for transactions and supports the script function that enables the card content to be modified. Before a transaction is completed, the card must remain inside of the reader.

This process may be applied via the contact and contactless interfaces. The standard debit/credit process via the contactless interface is mainly used for loading transactions of "Quick Pass" products.

2.3.1.2 qUICS

qUICS debit/credit process is generally briefed as qUICS process, which is designed to meet the requirements of quick payments. By optimizing the standard debit/credit command and transaction process, this process may shorten the time for transactions; however, it does not support the script function. The UICS allows the card holder to withdraw the card after the contactless reader has sensed the card while waiting for the completion of a transaction. It is the main application process for current "Quick Pass" products.

2.3.1.3 Low-value payment Process based on Debit/Credit Applications

This process is only applicable to E-Cash transactions via the contact interface.

2.3.2 Simplified Process for UnionPay Integrated Circuit Card

With respect to the refund, purchase cancellation, pre-authorization cancellation and pre-authorization completion cancellation transactions, the acquirer is not

required to send specific information about the card transaction, such as ARQC, to the issuer. Therefore, the interaction between the terminal and the card may be also simplified by adopting simplified process.

By adopting simplified process, the POS terminal may complete the follow-up process only based on the card number, card sequence number, term of validity and other data associated with the transaction read by the UnionPay Integrated Circuit Card, and does not need follow the standard debit/credit process. For application of the simplified process, please refer to *UnionPay Integrated Circuit Card Specifications—Basic Specifications Part 3 Debit/Credit Application Terminal Specification*.

3 Suggestions upon Transaction Path for UnionPay Integrated Circuit Card transactions via Different Transaction Channels

3.1 Transaction Path at POS Terminals

1. Debit/Credit Products

- (1) The contact debit/credit process via the contact interface shall be applicable to balance inquiry, purchase, pre-authorization and loading of debit/credit products.
- (2) The simplified process may be applicable to purchase cancellation, refund, preauthorization cancellation, and preauthorization completion cancellation transactions.

2. Quick Pass Products

- (1) The qUICS process shall be applicable to offline purchase, online balance inquiry, online purchase and pre-authorization transactions.
- (2) The contactless debit/credit process shall be applicable to loading transactions.

3. The others

The low-value payment process based on the debit/credit applications shall be applicable to contact E-Cash transactions.

3.2 Transaction Path at ATM and Self-service Terminals

1. Debit/Credit Products

- (1) The contact debit/credit process shall be applicable to inquiry, deposit, withdrawal, transferring, loading and unloading of debit/credit products.

2. Quick Pass Products

- (1) The qUICS process shall be applicable to online inquiry, deposit, withdrawal and transferring of quick pass products.
- (2) The contactless debit/credit process shall be applicable to loading transactions.

3.3 Suggestions upon Transaction Path at Internet Terminals

1. Debit/Credit Products

The contact debit/credit process shall be applicable.

2. Quick Pass Products

The contactless debit/credit process shall be applicable.

4 Requirements for Functions of Terminals and Cards

4.1 Functions Supported by Cards

1. The contact UnionPay Integrated Circuit Card shall be able to support the contact debit/credit process.
2. The contactless UnionPay Integrated Circuit Card shall be able to support the contactless debit/credit process (for loading transactions) and qUICS process.
3. The dual-interface UnionPay Integrated Circuit Card (contact and contactless interfaces) shall be able to support the debit/credit process and the qUICS process. The debit/credit process is suggested to support both the contact and contactless interfaces.

5 Suggestions upon Transaction Process at POS Terminals

To provide unified payment experiences to card holders, suggestions upon transaction process at POS terminals are presented as follows:

1. If both contact and contactless interfaces are available, the terminal will automatically select an interface according to the card presenting manner of the cardholder (inserting or waving the card).
2. If the UnionPay Integrated Circuit Card is used via the contact interface, the contact standard debit/credit process shall be followed preferentially to complete the online payment transaction. The terminal will not prompt the card holder to use the E-Cash function.
3. The contactless offline minimum amount shall be set at terminals. For a transaction with the value less than the minimum purchase amount, the qUICS transaction process shall be followed for offline purchase. If the offline E-cash balance of the card is insufficient and the terminal has the online capability, the transaction may be completed online following the qUICS process. With respect to purchase transactions with the value larger than the minimum amount, the qUICS process may be followed directly to complete online purchase.
4. If it is really necessary to use the contact E-Cash payment function for purchase, an E-Cash shortcut purchase key may be set at the POS terminal to enter the contact standard debit/credit process to complete the payment.

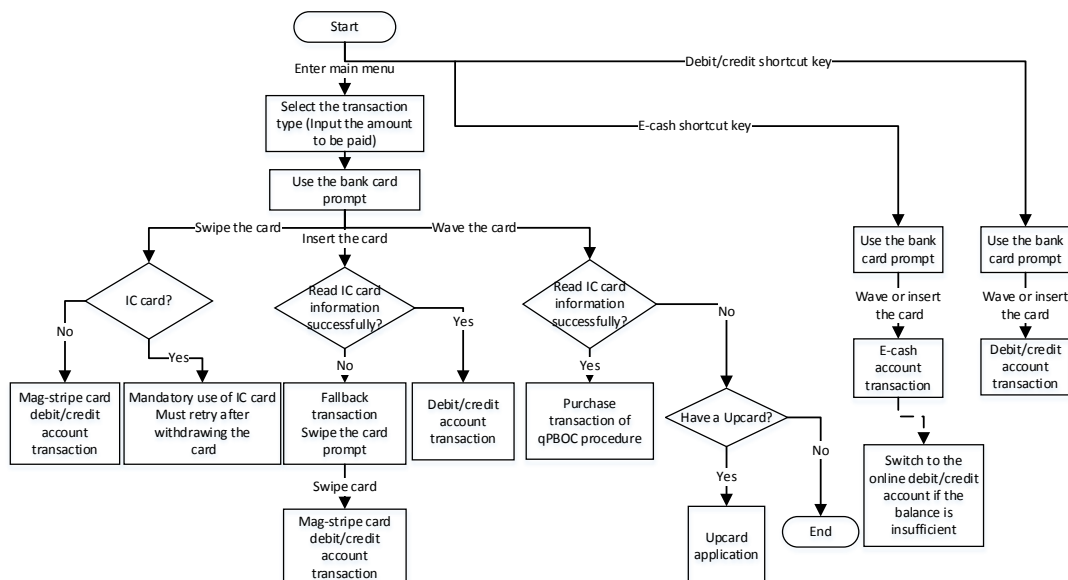


Figure 1 Suggested transaction flow at the POS terminal

6 Suggestions upon Processing Flow at ATM Terminals

1. If both contact and contactless interfaces are available, the terminal will automatically select an interface according to the card presenting manner of the cardholder (inserting or waving the card).
2. The ATM shall support UnionPay Integrated Circuit Card transactions that meet the *UnionPay Integrated Circuit Card Specifications*. The main processing flow is presented as follow; the basic processing requirements include but are not limited to:
 - The ATM terminal supports UnionPay Integrated Circuit Card and magstripe cards simultaneously. After the card is inserted, the ATM will first supply power to the chip.
 - If the power supply fails or the power supply succeeds but an error occurs during the terminal and the chip execute the application selection process as regulated in the *UnionPay Integrated Circuit Card Specifications* (the application jointly supported by the terminal and the card is unavailable), the terminal will read the magstripe and enter the magstripe processing process. If the service code in the magstripe is 2 or 6, the fallback message shall be filled out correctly.
 - If the power supply succeeds, and an application jointly supported by the terminal and card has been selected, but an error occurs during the terminal and chip execute the application initialization or during following processes as regulated in the *UnionPay Integrated Circuit Card Specifications*, the transaction will be terminated and the card will be returned directly.
3. For ATM terminals that accept transactions via contactless UnionPay Integrated Circuit Card/mobiles, the terminal operation interface may be started and entered by any means.
 - The contactless reader shall be always in the activation and working state when on standby and shall initiate the operation interface after detecting a card has effectively entered the induction area; it is suggested that the acceptor set induction mechanisms (such as the card holder's approach induction and the random key induction). The contactless reader may be closed when on standby while there are no card holders nearby.
 - If the contactless reader is closed when on standby, corresponding options shall be provided at the terminal. When the card holder chooses to conduct transactions via the contactless means, the contactless reader will start and enter into corresponding operation interface.

The second type is adopted in processes presented below.

The UnionPay Integrated Circuit Card reading process includes the power supply and application selections, refer to relevant parts of the *UnionPay Integrated Circuit Card Specifications* for details.

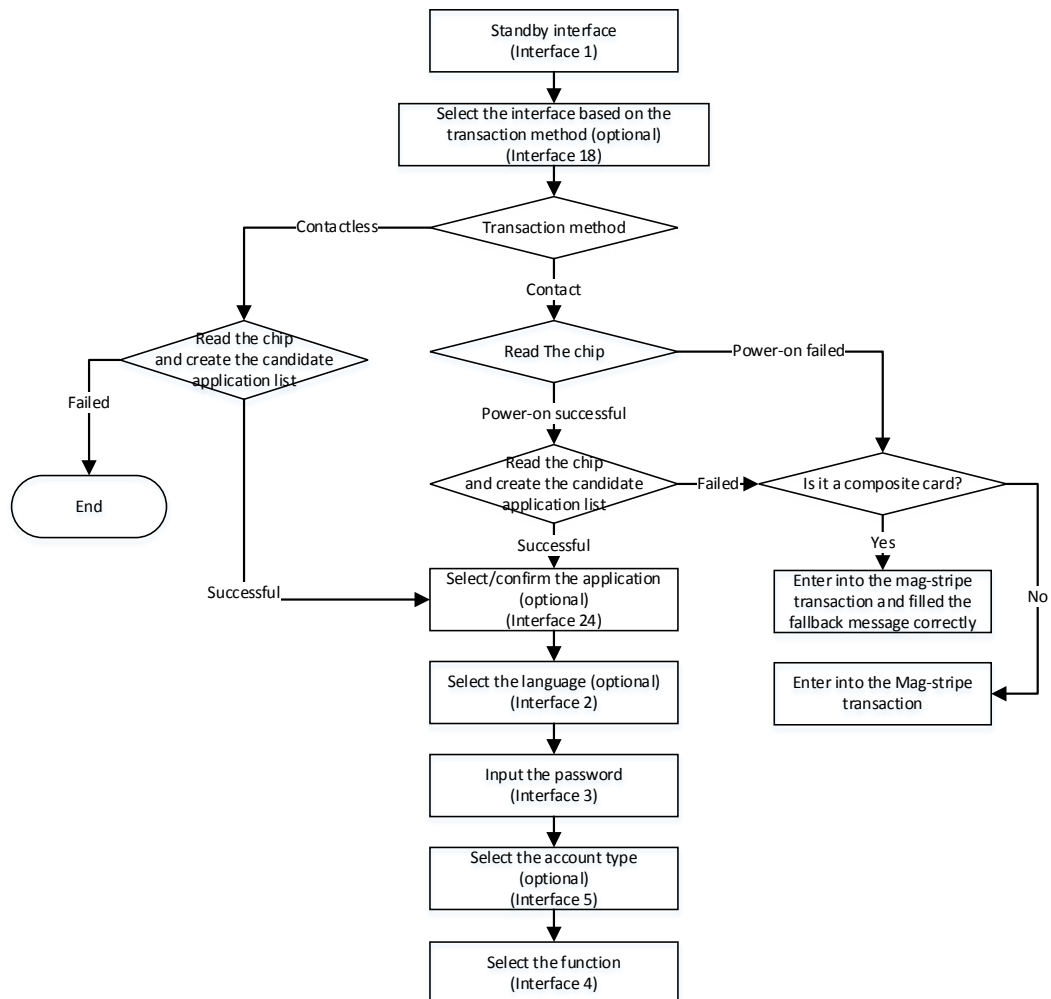


Figure 2 Suggested card processing flow at the ATM terminal